Vibration Control of Footbridges Based on Local Resonance Band Gaps

Linyun Zhou[1] and Shui Wan[2]

Abstract: As footbridges are lightweight and slender structures, they are highly sensitive to human-induced excitation, and control measures are required to suppress undesirable structural vibrations. As wave propagation is improbable within a band gap, this study introduces a band gap in a footbridge and develops a new technique as an alternative to suppress the human-induced vibration response. Inerter-based dynamic vibration absorbers (IDVAs) are arranged periodically in the footbridge, which can convert the conventional box girder into a specially designed periodic metamaterial beam with a local resonance band gap. Following the spectral element method, the band gap structure of the metamaterial beam with IDVAs is proposed and validated by test results and numerical experiments. The band gap structure can be simultaneously tuned by the parameters of the IDVAs, including spring stiffness, inertance, and attached mass. A computer-based program was presented to determine the reasonable design parameters of IDVAs. Finally, a new vibration attenuation method for footbridges is proposed based on the theory of metamaterials and validated by numerical experiments. The results show that the proposed method exhibits good performance in vibration attenuation. **DOI: 10.1061/(ASCE)ST.1943-541X.0003454.**

Author keywords: Footbridges; Metamaterial beam; Band gap; Vibration reduction; Periodic structures.

Introduction

With the rapid development of high-performance and lightweight structural materials, footbridges have gradually become longer, lighter, and slender based on aesthetic requirements. As a result, the low natural frequencies of the structure may be within the range of the pedestrian frequency, and an excessive vibration response is likely to occur that exceeds the walking comfort requirement (Zivanovic et al. 2005; Caetano et al. 2010). The vibration induced by pedestrian excitation is generally considered a serviceability problem rather than a strength problem, and the deformation of the structures is quite low compared to the predetermined limit value (Jones et al. 1981). Thus, the vibration acceleration of footbridges will be the predominant factor in structural design. To date, limitations on vibration acceleration have been suggested based on the human comfort criterion in design codes such as Eurocode 5 (CEN 2005), AFGC (2006), ISO 10137 (ISO 2007), and HIVOSS (Feldmann et al. 2008).

Conventional vibration mitigation techniques for footbridges are based on the structural dynamic theory. A tuned mass damper (TMD), which operates by damping the vibration system to dissipate the energy, has been widely adopted to reduce the vibration response in engineering structures (Ormondroyd and Hartog 1928). However, the control efficiency of a single TMD system highly depends on the frequency; the frequency is equal to or very close to the structural natural frequency. Moreover, the detuning problem leads to poor mitigation performance of TMDs (Werkle et al. 2013). Recently, a robust TMD optimization approach has been developed and used in footbridges by accounting for the uncertainties in the loading and modal parameters of the main structure (Lievens et al. 2016; Shi et al. 2019). To widen the controlling frequency and improve the efficiency, Poovarodom et al. (2003) adopted a multiple-tuned mass damper (MTMD) to control the human-induced vibrations in a footbridge, and the control effectiveness of the MTMD was investigated based on a single pedestrian excitation. Subsequently, considerable efforts were made to mitigate the dynamic response of footbridges based on MTMD, emphasizing the robustness of its control performance due to the off-tuning effect (Costa et al. 2021; Zhu et al. 2021). The effectiveness and applicability of MTMDs depend heavily on the attached mass. The larger the attached mass, the better the vibration suppression and robustness to detuning. However, a heavy attached mass in superstructure box girders is not allowed owing to economic and structural limitations. To enhance TMD vibration suppression effectiveness, Angelis et al. (2021) incorporated an inerter into a conventional grounded TMD to mitigate the dynamic response of a footbridge. Inerters were proposed by Smith (2002) as two-terminal mechanical elements in which the forces applied are proportional to the relative acceleration at the nodes. Various physical models have been proposed and can be classified into two main types, the flywheel-based and the fluid-based (Hu and Chen 2015; Liu et al. 2018; Taflanidis et al. 2019; Pietrosanti et al. 2020). Compared to the traditional TMD, the inerter-based TMD has a small or negligible mass, which can significantly reduce the vibration response (Wen et al. 2016). Although many inerter-based suspension configurations were proposed and used in bridge engineering (Dai et al. 2019), tall buildings (Petrini et al. 2020), and earthquake engineering (Kaveh et al. 2020), very few efforts have been made to control footbridge vibration using an inerter-based TMD.

Elastic metamaterials are periodic structures in which the local resonators are periodically arranged (Kushwaha et al. 1993). The most prominent dynamic characteristic of the metamaterials is that they possess frequency pass-bands and band gaps (Ma and Sheng 2016). The elastic waves attenuate significantly in the band gaps, and this property is potentially used for vibration control in engineering. Liu et al. (2000) first demonstrated local resonance (LR)

[1]Postdoctoral, School of Transportation, Southeast Univ., 2 Sipailou, Nanjing 210096, China (corresponding author). Email: 101300072@seu.edu.cn

[2]Professor, School of Transportation, Southeast Univ., 2 Sipailou, Nanjing 210096, China. Email: 421681743@qq.com

Note. This manuscript was submitted on July 29, 2021; approved on May 10, 2022; published online on July 9, 2022. Discussion period open until December 9, 2022; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Structural Engineering***, © ASCE, ISSN 0733-9445.

acoustic metamaterials that display band gaps at lattice scales two orders of magnitude smaller than the wavelength. Subsequently, beams with periodically attached resonators (i.e., LR beams) have attracted increasing attention for vibration attenuation due to low-frequency band gaps (Failla et al. 2020). Locally resonant units play a dominant role in blocking wave transmission. As a result, heavy resonators are needed to open a band gap at low frequencies. To achieve low-frequency band gaps with a small mass, Yilmaz et al. (2007) introduced inertial amplification mechanisms into acoustic metamaterials. The inertia of a resonating mass was magnified to a degree proportional to the arm length, and thus, large inertial forces were generated by amplifying the motion of a small mass, reducing its resonance frequency correspondingly (Yilmaz and Hulbert 2010). Frandsen et al. (2016) studied the longitudinal wave characteristics of a continuous rod with a periodic array of inertial amplification mechanisms. The inertial mechanism band gap can be realized with almost 20 times less added mass compared to that needed in a standard local resonator. Thereafter, substantial efforts were made to achieve low-frequency band gaps using inerter-based absorbers for metamaterial chains (Banerjee et al. 2021), rods (Kulkarni and Manimala 2016), lattice (Yuksel and Yilmaz 2020), and Euler–Bernoulli beams with rectangular cross sections (Li and Li 2018; Fang et al. 2018). However, no attempts have been made to investigate metamaterial beams with box sections, which are widely used in civil engineering. Moreover, it should be noted that compared with the vibration frequency of a footbridge, the band gaps in all previous studies were quite high (at least 100 Hz) and the total attached mass was quite heavy (more than 60% of the host structure).

Inspired by the unique physical property of metamaterial beams, that is, vibration waves are impeded within the band gap, this study introduces a band gap into a footbridge to mitigate the human-induced vibration response. However, obtaining a low-frequency band gap in conventional box girders while ensuring that the width of the band gap covers the full frequency range of pedestrian excitation remains a challenge. To achieve the low-frequency band gap with a small attached mass, inerter-based dynamic vibration absorbers (IDVAs) are arranged periodically in the box girder, which can convert the traditional box girder into a metamaterial beam with an LR band gap. It should be noted that the IDVAs are considered within the framework of periodic structures, that is, they are used only to generate the band gap. The spectral element method (SEM) is adopted to study the band gap structure of the metamaterial beam and validated through numerical experiments and test results. The effect of IDVA design parameters on the band gap is analyzed. Further, a new technique for footbridge vibration mitigation is proposed. The results demonstrate that the vibration response can be suppressed effectively in all load cases when the pedestrian frequency falls in the band gap.

Band Gap Structure of Timoshenko Beam with IDVAs

Fig. 1 shows a simple model of an LR beam. IDVAs are periodically attached to a Timoshenko beam; as a result, this conventional beam is converted into a metamaterial beam with a band gap structure that depends on the parameters of IDVAs. The spacing of IDVAs is denoted by L, which is the length of the unit cell. The vibration governing equation of the beam without IDVAs is given as

$$\kappa GS\left[\frac{\partial^2 w(x,t)}{\partial x^2}-\frac{\partial\theta(x,t)}{\partial x}\right]-\rho S\frac{\partial^2 w(x,t)}{\partial t^2}=0 \qquad (1a)$$

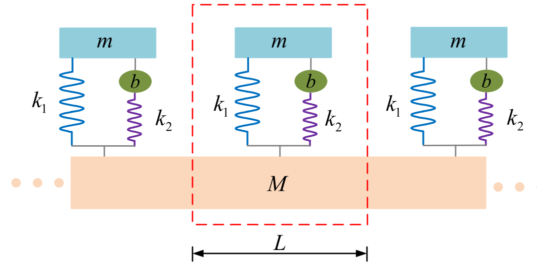

Fig. 1. LR Timoshenko beam with IDVAs.

$$EI\frac{\partial\theta^2(x,t)}{\partial x^2}+\kappa GS\left[\frac{\partial w(x,t)}{\partial x}-\theta(x,t)\right]-\rho I\frac{\partial^2\theta(x,t)}{\partial t^2}=0 \quad (1b)$$

where w = deformation of the beam; θ = rotational angle; E and G = Young's modulus and shear modulus, respectively; κ = shearing-shape coefficient; ρ = mass density; and S and I = area and inertia of the beam cross section, respectively.

According to the theory of elasticity, the moment and shear force can be obtained by

$$M(x,t)=EI\frac{\partial\theta(x,t)}{\partial x} \qquad (2a)$$

$$Q(x,t)=\kappa GS\left[\frac{\partial w(x,t)}{\partial x}-\theta(x,t)\right] \qquad (2b)$$

The steady-state solution for the governing Eq. (1) can be written in the form

$$w(x,t)=\frac{1}{N}\sum_{n=0}^{N-1}W_n(x,\omega_n)e^{i\omega_n t} \qquad (3a)$$

$$\theta(x,t)=\frac{1}{N}\sum_{n=0}^{N-1}\psi_n(x,\omega_n)e^{i\omega_n t} \qquad (3b)$$

where ω_n = frequency; and W_n and ψ_n = spectral components of the generalized displacements, respectively.

Substituting Eq. (3) into Eq. (1) yields

$$\kappa GS[W''(x,\omega)-\psi(x,\omega)]+\rho S\omega^2 W(x,\omega)=0 \qquad (4a)$$

$$EI\psi''(x,\omega)+\kappa GS[W'(x,\omega)-\psi(x,\omega)]+\rho I\omega^2\psi(x,\omega)=0 \qquad (4b)$$

It is assumed that the general solution of the presented equation can be expressed as

$$W(x,\omega)=ae^{-ik(\omega)t} \qquad (5a)$$

$$\psi(x,\omega)=\beta ae^{-ik(\omega)t} \qquad 5(b)$$

Substituting Eq. (5) into Eq. (4), we have

$$\begin{bmatrix}\kappa GSk^2-\rho S\omega^2 & -ik\kappa GS\\ ik\kappa GS & EIk^2+\kappa GS-\rho I\omega^2\end{bmatrix}\begin{Bmatrix}1\\ \beta\end{Bmatrix}=\begin{Bmatrix}0\\ 0\end{Bmatrix} \qquad (6)$$

Thus, the dispersion equation can be expressed by

$$k^4-\eta_1 k_F^4 k^2-k_F^4(1-\eta_2 k_G^4)=0 \qquad (7)$$

in which, the coefficients are given by

J. Struct. Eng., 2022, 148(9): 04022137

$$k_F = (\omega^2 \rho S/EI)^{1/4} \tag{8a}$$

$$k_G = (\omega^2 \rho/\kappa G)^{1/4} \tag{8b}$$

$$\eta_1 = \frac{S}{I} + \frac{EI}{\kappa GS} \tag{8c}$$

$$\eta_2 = \frac{S}{I} \tag{8d}$$

The four roots of Eq. (7) can be derived by

$$k_1 = -k_2 = \frac{1}{\sqrt{2}} k_F \sqrt{\eta_1 k_F^2 + \sqrt{\eta_1^2 k_F^4 + 4(1-\eta_2 k_G^4)}} \equiv k_t \tag{9a}$$

$$k_3 = -k_4 = \frac{1}{\sqrt{2}} k_F \sqrt{\eta_1 k_F^2 - \sqrt{\eta_1^2 k_F^4 + 4(1-\eta_2 k_G^4)}} \equiv k_e \tag{9b}$$

According to the SEM, the spectral matrix, $S_T(\omega)$, can be given by

$$S_T(\omega) = S_T(\omega)^{\mathrm{T}} = EI \begin{bmatrix} s_{11} & s_{12} & s_{13} & s_{14} \\ s_{21} & s_{22} & s_{23} & s_{24} \\ s_{31} & s_{32} & s_{33} & s_{34} \\ s_{41} & s_{42} & s_{43} & s_{44} \end{bmatrix} \tag{10}$$

where the matrix elements are determined by

$$\begin{aligned} s_{11} = s_{33} = iD_1 D_3[(e_t^2-1)(e_e^2+1)r_t \\ -(e_t^2+1)(e_e^2-1)r_e](k_t r_t - k_e r_e) \end{aligned} \tag{11a}$$

$$\begin{aligned} s_{12} &= -s_{34} \\ &= D_1 D_3[(e_t^2-1)(e_e^2-1)(k_t r_e + k_e r_t) \\ &\quad -(e_t^2+1)(e_e^2+1)(k_t r_t + k_e r_e)] \end{aligned} \tag{11b}$$

$$s_{13} = iD_1 D_3[(e_t^2-1)e_e r_t - (e_e^2-1)e_t r_e](k_e r_e - k_t r_t) \tag{11c}$$

$$s_{14} = -s_{23} = -2D_1 D_3 (e_t - e_e)(1 - e_t e_e)(k_e r_e - k_t r_t) \tag{11d}$$

$$s_{22} = s_{44} = D_2 D_3[(e_t^2-1)(e_e^2+1)r_e - (e_t^2+1)(e_e^2-1)r_t] \tag{11e}$$

$$s_{24} = D_2 D_3[(e_e^2-1)e_t r_t - (e_t^2-1)e_e r_e] \tag{11f}$$

in which

$$e_t = e^{-ik_t L} \tag{12a}$$

$$e_e = e^{-ik_e L} \tag{12b}$$

$$r_t = (k_t^2 - k_G^4)/k_t \tag{12c}$$

$$r_e = (k_e^2 - k_G^4)/k_e \tag{12d}$$

$$D_1 = \frac{1}{8}\left(1 - \eta_1 \frac{I}{S} k_G^4\right)^{-1/2} \tag{12e}$$

$$D_2 = \frac{1}{8}\left[\left(\eta_1 \frac{I}{S} + \frac{EI}{\kappa GS}\right)^2 k_F^4 + 4\left(1 - \eta_1 \frac{I}{S} k_G^4\right)\right]^{1/2} \tag{12f}$$

$$\begin{aligned} D_3 = -8ik_F^2[(2r_t r_e(e_t^2+1)(e_e^2+1) - 4e_t e_e \\ -(r_t^2+r_e^2)(e_t^2-1)(e_e^2-1)] \end{aligned} \tag{12g}$$

According to Fang et al. (2018), the effective dynamic matrix for the IDVA is

$$S_{\mathrm{IDVA}}(\omega) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & F_d(\omega) & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{13}$$

where

$$F_d(\omega) = \frac{-m_1\omega^2}{1 - m_1\omega^2/[G'(\omega) + m_1\omega^2]} \tag{14}$$

in which, the nodal frequency response function $G'(\omega)$ for IDVA can be rewritten as

$$G'(\omega) = k_1 + \frac{1}{1/k_2 + 1/(ic\omega - b\omega^2)} - m_1\omega^2 \tag{15}$$

where k_1 = stiffness of the spring connecting the beam element and absorber; k_2 = stiffness of the spring connecting the inerter and beam element; b = inertance; m_1 and c = mass and damping of the absorber, respectively. For convenience, nondimensional coefficients are defined in the calculation as damping ratio $\xi = c/2\sqrt{m_1 k_1}$ and inertance-to-mass ratio $b' = b/m_1$.

Then, the force equation of the jth unit cell in the frequency domain can be derived as

$$F_j(\omega) = \begin{bmatrix} -Q_j(0,\omega) \\ -M_j(0,\omega) \\ Q_j(L,\omega) + F_d(\omega) \\ M_j(L,\omega) \end{bmatrix} \tag{16}$$

According to the Floquet boundary condition for the beam element, the spectral components of displacement, rotation angle, shear forces, and bending moments at the ends of the element ($x = 0$ and $x = L$) satisfy

$$\begin{Bmatrix} W_j(L,\omega) \\ \psi_j(L,\omega) \\ Q_j(L,\omega) \\ M_j(L,\omega) \end{Bmatrix} = e^{\mu} \begin{Bmatrix} W_j(0,\omega) \\ \psi_j(0,\omega) \\ -Q_j(0,\omega) \\ -M_j(0,\omega) \end{Bmatrix} \tag{17}$$

Substituting Eq. (17) into the spectral nodal force in Eq. (16), the eigenvalue problem is obtained as follows:

$$[S_{12}e^{\mu} + (S_{11} + S_A(\omega))e^{\mu} + S_{22}e^{\mu} + S_{21}e^{2\mu}] \begin{bmatrix} W_j(0,\omega) \\ \psi_j(0,\omega) \end{bmatrix} = 0 \tag{18}$$

where μ = wave propagation constant, and $S_A(\omega)$ is defined as

$$S_A(\omega) = \begin{bmatrix} F_d(\omega) & 0 \\ 0 & 0 \end{bmatrix} \tag{19}$$

The wave propagation constant μ and the corresponding band gap structure of an infinite metamaterial beam can be obtained by solving the eigenvalue problem determined by Eq. (18). The real part of μ represents the attenuation of the wave. In the band gap, the real part of the wave propagation constant is nonzero and no wave is allowed to propagate.

For the finite metamaterial beam with N unit cells, the global equilibrium equation can be derived by assembling all the spectral element matrix and IDVAs as follows:

$$\sum_{j=1}^{N}[S_{Tj}(\omega)S_{Aj}(\omega)]d_j = \sum_{j=1}^{N}F_j(\omega) \tag{20}$$

where $\sum_{j=1}^{N} F_j(\omega)$ = external load.

Then, the displacement transmission of the metamaterial beam can be given by

$$T(\omega) = 20\log_{10}\left|\frac{W_N(L,\omega)}{W_1(0,\omega)}\right| \tag{21}$$

where $W_1(0,\omega)$ and $W_N(L,\omega)$ = displacements at the excitation point and at the other end of the finite metamaterial beam, respectively.

Band Gap of Box Girder

A cantilever box girder with IDVAs was designed to validate the proposed band gap of the Timoshenko beam. The inertance ratio was set as $b' = 1.0$. Local resonators were arranged periodically on the bottom slab as shown in Fig. 2. The beam with a length of 1.2 m was made of polymethyl methacrylate (PMMA). The Young's modulus, mass density, and passion ratio were set to be $E = 3 \times 10^9$ MPa, $\rho = 1,142\ \mathrm{kg/m^3}$, and 0.38, respectively. The widths of the top flange and bottom slab were 0.1 m and 0.045 m, respectively, and the height of the beams was $h = 0.065$ m. The thickness of the top flange and bottom slab was 0.005 m. Ten spring-mass resonators were attached to the bottom slab with equal spacing, i.e., the unit cell is 0.1 m. The spring stiffness was set to $k_1 = k_2 = 2 \times 10^4$ N/m; the mass of the resonator was 0.1 kg. A sinusoidal excitation with constant displacement was applied to the beam at a distance of 0.3 m from the fixed end. Two accelerometers were placed at the excitation point and the free end of the beam, respectively, to monitor the vibration acceleration response. Through the sine sweep test and fast Fourier transform (FFT), the vertical acceleration transmission of the box girder with local resonators can be illustrated as shown in Fig. 3. Fig. 4 shows the vertical acceleration responses of the traditional and metamaterial beams under different vibration frequencies $f = 100$ Hz (pass band) and $f = 125$ Hz (band gap). It can be observed that the vertical acceleration at the free end is higher than that at the excitation point when the frequency of excitation lies in the pass band ($f = 100$ Hz). This is because the number of resonance peaks increases owing to the attached local resonators, leading to an amplification of the vibration response. When the excitation frequency is in the band gap ($f = 125$ Hz), the acceleration at the free end is significantly lower than that at the excitation point. This implies that the vibration response can be attenuated effectively when the excitation frequency falls in the band gap.

Based on the principle of equal section area and inertia, the box girder can be simplified as an equivalent Timoshenko beam with a width of 0.015 m and a height of 0.095 m. According to Eq. (22), the starting and cutoff frequencies of the LR1 band gap are $f_{s,1} = 47.2$ Hz and $f_{c,1} = 50.4$ Hz, respectively, and the corresponding frequencies of LR2 are $f_{s,2} = 119.2$ Hz and $f_{c,2} = 129.4$ Hz. It can be observed that the calculations are consistent with the test results. The relative errors of the starting frequencies of LR1 and LR2 are only 1.67% and 1.18%, respectively, and those of the cutoff frequencies of LR1 and LR2 are 2.15% and 0.76%, respectively. Moreover, the acceleration transmission curve based on Eq. (25) is consistent with the test curve. Thus, the proposed method can be used to analyze the band gap structure of the Timoshenko beam with IDVAs. As the transmission curve was developed based on the Euler beam (Xiao et al. 2012), the starting frequencies of LR1 and LR2 are slightly lower than those of the test results, and the relative errors are only 0.86% and 1.44%, respectively. However, the cutoff frequency of LR2 is considerably larger than that of the test result, and the relative deviation reaches up to 11.5%. As a result, the bandwidth of LR2 is approximately 2.5 times that of the test result. In other words, the band gap structure developed based on the Euler

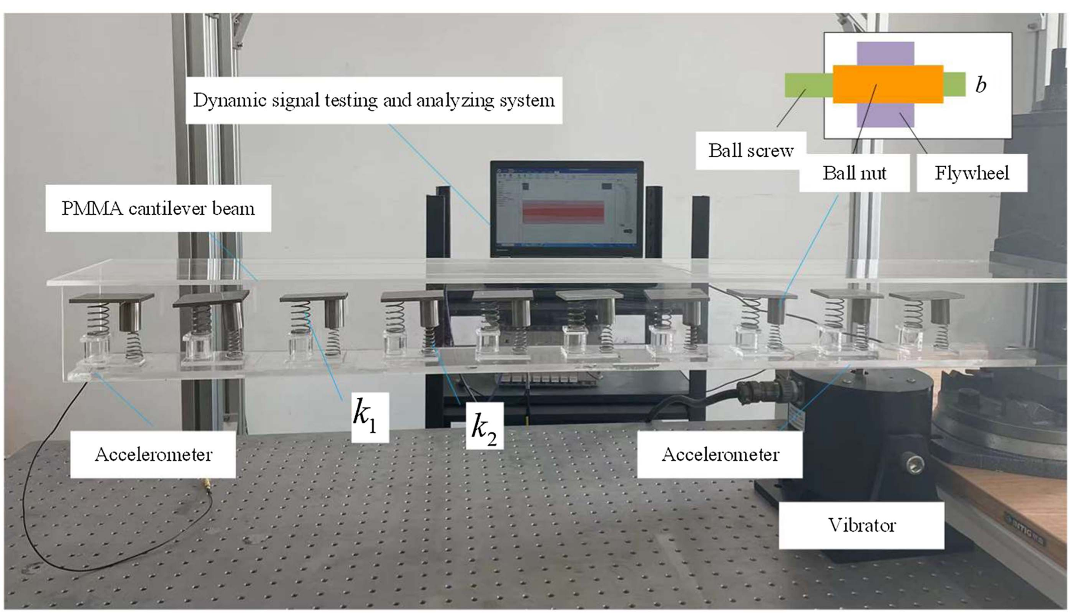

Fig. 2. Cantilever beam with locally spring-mass resonators.

J. Struct. Eng., 2022, 148(9): 04022137

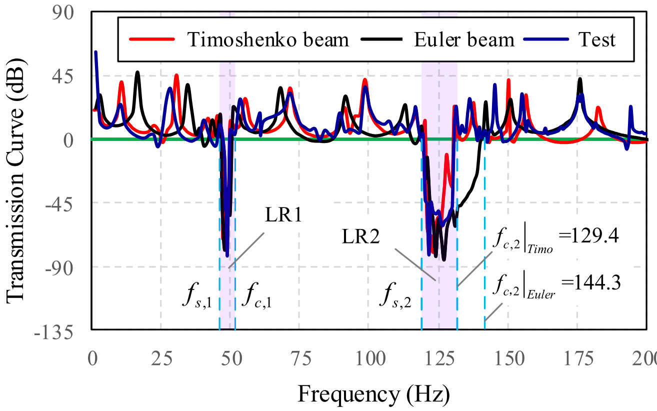

Fig. 3. Transmission curve for LR beams.

beam overestimated the bandwidth, which is unreasonable for the beams in civil engineering.

To further validate the band gap of the box girder with IDVAs, a finite element model (FEM) of the box girder reported by Wang et al. (2019) was built. The clear span and height of the beam were 64 m and 2.5 m, respectively. The width of the deck was 5.4 m. Detailed dimensions of the cross section are illustrated in Fig. 5. The thickness of the shaped steel plate of the steel box girder was 16 mm. The elastic modulus and Poisson ratio of the steel were set as $E = 2.1 \times 10^{11}$ MPa and 0.3, respectively. The area and inertia of the cross section were 0.3628 m^2 and 0.3576 m^4, respectively. Based on the principle of equal cross-sectional area and inertia, the height and width of the equivalent rectangular cross section were obtained as 3.44 m and 0.11 m, respectively. The length of the unit cell was assumed to be $L = .5$ m. Parameters of the IDVAs were set as $k_1 = 2.71 \times 10^5$ N/m, $k_2 = 8.54 \times 10^4$ N/m, $m = 221$ kg, and $b' = 33$.

Table 1 shows the first eight frequencies of the equivalent beam. It can be observed that the natural frequencies of the equivalent beam are consistent with those of the real structure. Fig. 6 shows a comparison of the transmission curves of the FEM and equivalent beam model. It can be observed that there is a discrepancy between the two models because the equivalent rectangular beam model could not accurately represent the behavior of the box girder. However, the transmission curves of the two models with different damping ratios are observed to be consistent. Most importantly, the frequency of the band gap obtained by the equivalent rectangular beam model is similar to that of the FEM. Thus, the equivalent beam model can be used to design the band gap of the box girders.

Design of Band Gap Structure

The LR band gap structure of a metamaterial beam mainly depends on the natural frequency of the IDVAs (Xiao et al. 2012), which is governed by the attached mass, spring stiffness, and inertance. An infinite metamaterial beam with a rectangular cross section of $0.1\ \mathrm{m} \times 0.1\ \mathrm{m}$ was considered to investigate the propagation behavior of flexural waves and the tunability of band gaps by adjusting the parameters of IDVAs. The length of the unit cell was set as $L = 1.0$ m. The Young's modulus and mass density were $E = 2.1 \times 10^{11}$ MPa and $\rho = 7850\ \mathrm{kg/m^3}$, respectively.

First, the parameters of IDVAs were set as $m_1 = 7.8$ kg, $k_1 = 2 \times 10^3$ N/m, $k_2 = 2.4 \times 10^3$ N/m, and $b' = 5$, respectively. The mass ratio is defined as $m' = m_1/(\rho A L)$, where ρ and A are the density and area of the cross section of the host beam, respectively. For comparison, the band gap structure of the metamaterial beam carrying simple spring-mass resonators (SSMRs) is plotted in Fig. 7, with the same starting frequency and spring stiffness as those in IDVAs. It is observed that the IDVA-based LR band gap is split into one narrow band gap at a lower frequency (LR1) (1.44–1.72 Hz) and one broad band gap at a higher frequency (LR2) (3.76–4.24 Hz). Moreover, to generate the same starting frequency, the attached mass of SSMRs is required to be 25 times that of IDVAs. Thus, the IDVAs can lower the band frequency and broaden the total bandwidth with a lightweight attached mass. Fig. 8 shows the band gap map with different attached mass ratios. The solid lines and dashed lines represent the cutoff frequency and starting frequency of LR1 (blue line) and LR2 (red line), respectively. It can be observed that the edge frequency of LR1 and LR2 gradually decreases with the increased attached mass ratios, and the bandwidth decreases rapidly with increasing attached mass.

Fig. 9 shows the wave propagation constant of the metamaterial beam with varying inertance-to-mass ratio b' and constant $m' = 0.1$, $k_1 = 2 \times 10^3$ N/m, and $k_2 = 2.4 \times 10^3$ N/m. It can be observed that the starting frequency of LR1 decreases slowly with b', whereas the starting frequency of LR2 decreases quickly with increasing b'. As b' increases, the width of LR2 with high frequency gradually increases and the width of LR1 with low frequency gradually decreases. This indicates that the energy in the

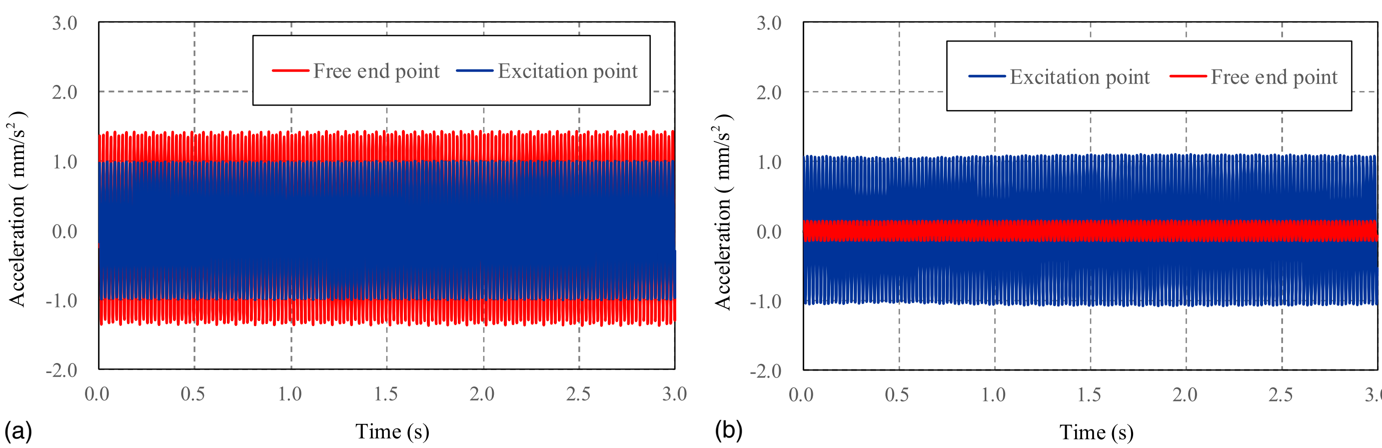

Fig. 4. Acceleration response of the metamaterial beam under different excitation frequencies: (a) $f = 100$ Hz; and (b) $f = 125$ Hz.

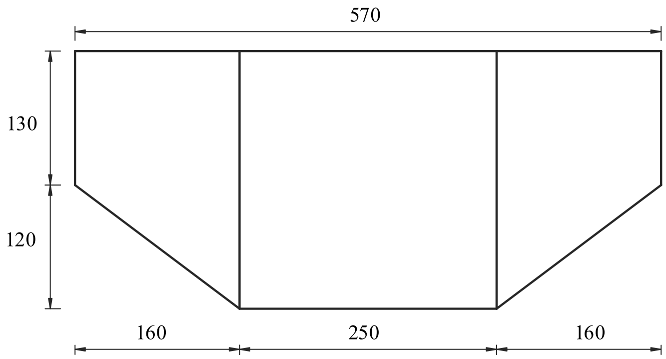

Fig. 5. Cross section of the steel footbridge (dimension: mm).

Table 1. Frequency of the footbridge and equivalent Timoshenko beam

Mode number	Frequency: Test	Frequency: Wang et al. (2019)	Frequency: Equivalent model	Relative error	Description
1	2.943	3.016	2.952	0.31%	Vertical bending
2	—	3.665	3.558	—	Lateral bending
3	9.372	9.733	9.411	0.42%	Vertical bending
4	—	11.414	11.172	—	Lateral bending
5	16.995	17.6	17.114	0.70%	Vertical bending
6	—	18.03	17.849	—	Longitudinal bending
7	22.692	23.41	22.961	1.25%	Vertical/bending
8	—	28.25	27.883	—	Lateral/bending

IDVA is redistributed, where the contribution of the inerters is dominant.

Fig. 10 shows the effect of k_1 on the band gap structure with $k_2 = 2.4 \times 10^3$ N/m, $m_1 = 7.8$ kg, and $b' = 5$. It can be observed that the starting frequency of LR2 (blue line) remains constant when $k_1/k_2 \leq 5$. Meanwhile, the other edge frequency bands increase gradually with increasing k_1. Fig. 11 shows the band map of the LR beam with constant $k_1 = 2 \times 10^3$ N/m, $m_1 = 7.8$ kg, and $b' = 10$. It can be seen that the starting frequency of LR1 remains almost constant with different spring stiffness k_2. Meanwhile, the pass band between LR1 and LR2 decreases gradually with decreasing k_2. As a result, the band gap near-coupling occurred when $0.05 < k_2/k_1 < 0.125$, that is, LR1 and LR2 were merged to be a new pseudo gap (LR1-2) with a band width of 0.85 Hz–3.05 Hz. Thus, band gap merging is an effective method to broaden the width of the band gap.

Although the band gap structure of the metamaterial Timoshenko beam can be tuned easily by adjusting the parameters of IDVAs, it is still difficult to determine the IDVA parameters. To simplify the design process and obtain the optimal solution governed by engineering requirements, a computer-based program is presented to obtain the optimal solution of the design parameters automatically, as shown in Fig. 12. The main steps in the process are described as follows:

1. Determine the range of excitation frequency based on the database or field test, and the designed starting frequency of the band gap should be less than the minimum excitation frequency f_{real}.
2. Choose a value for the unit cell length L.
3. Assume the value of m_1. The total mass of IDVAs should not exceed the specified limit due to economic and structural limitations.
4. Choose values of k_1 and the coefficient k_1/k_2.
5. Assume that the starting frequency is equal to f_{real} and calculate the value of b' according to Eq. (20).
6. Decrease k_2 until LR1 merges with LR2.
7. Check the compression length of the spring; if the compression length exceeds the permissible limits, go to step 4 and adjust the value of k_1; else, proceed to the next step.
8. Estimate the width of the merged band gap. If the bandwidth covers the range of the excitation frequency, output the parameter values and go to the next step; if not, go to step 3 and reassume the value of m_1.
9. Increase the value of L, and repeat the steps from 3 to 9. The process terminates when no real solution satisfies the requirements.

Vibration Reduction in the Footbridge

The footbridge used in a previous study (Wang et al. 2019) was adopted to illustrate the effectiveness of vibration attenuation by metamaterial band gaps. The vertical pedestrian excitation frequency ranges from 1.2 Hz to 3.2 Hz (Cheng and Hua 2009), that is, the designed starting band gap frequency is 1.2 Hz, and the width of the band gap should cover this range. Following the proposed design methods of metamaterial beams, the unique parameters of IDVAs obtained are listed in Table 2. It can be observed that when the length of the unit cell was 2.0 m, the width of the band

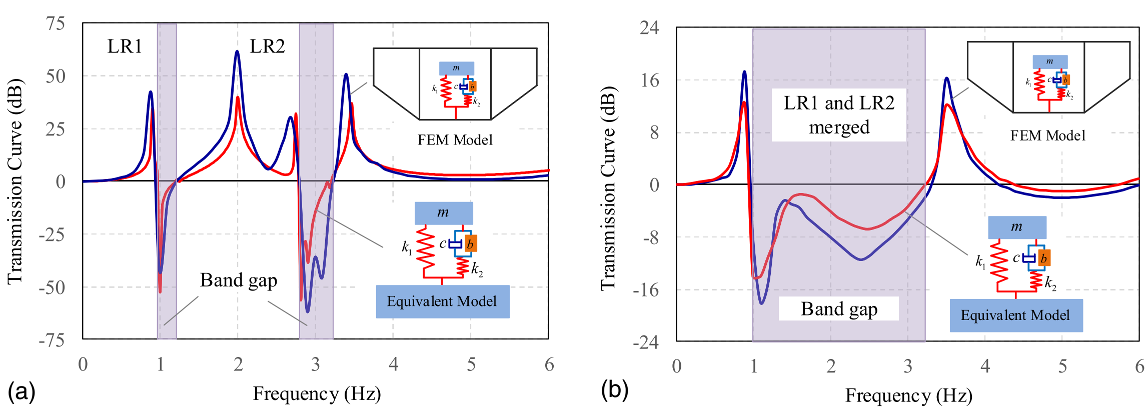

Fig. 6. Displacement transmission curve for different damping ratios: (a) $\xi = 0$; and (b) $\xi = 0.2$.

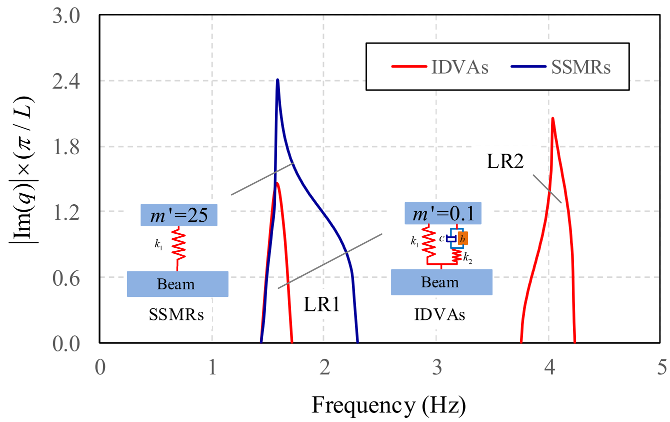

Fig. 7. Band gap of metamaterial beam with different dynamic absorbers.

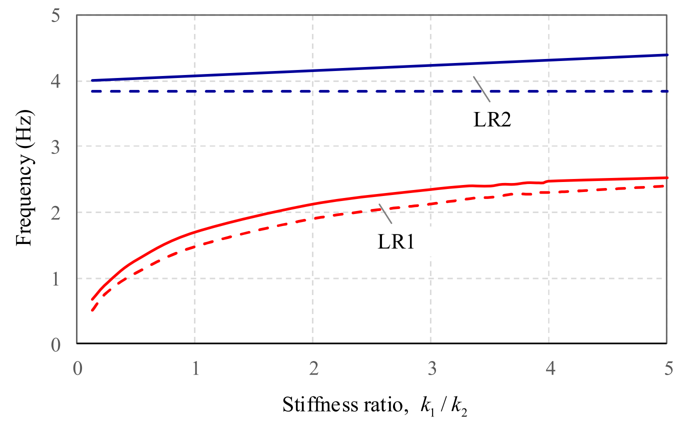

Fig. 10. Band gap behavior with different spring stiffness k_1.

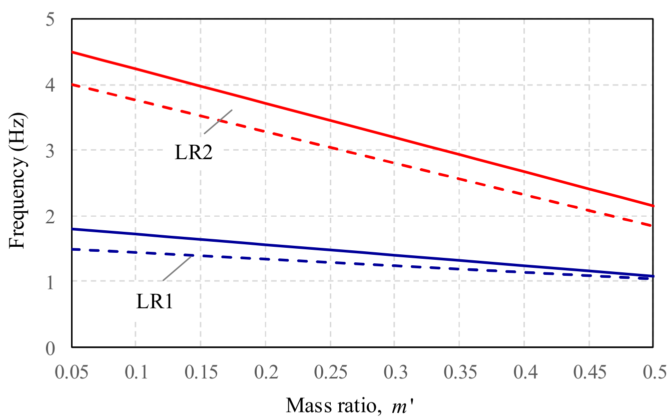

Fig. 8. Band gap map with different attached mass ratios of IDVAs.

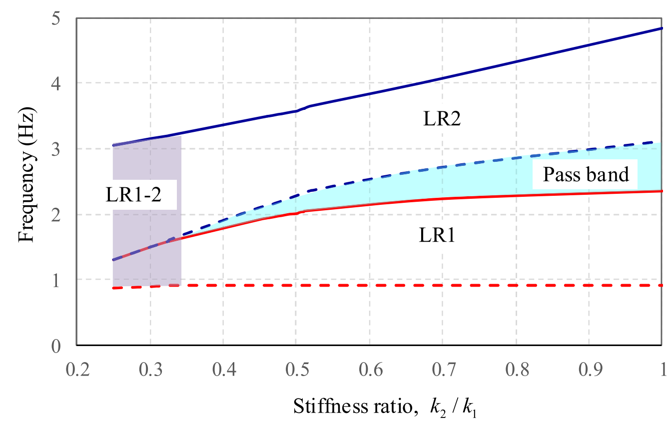

Fig. 11. Effect of spring stiffness k_2 on band gap.

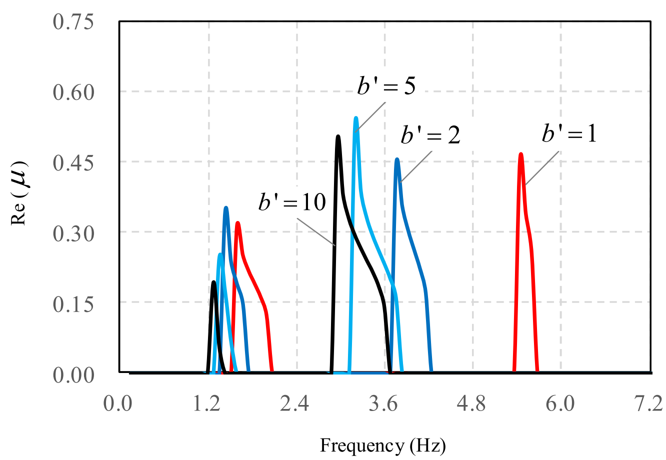

Fig. 9. Real part of propagation constant versus frequency.

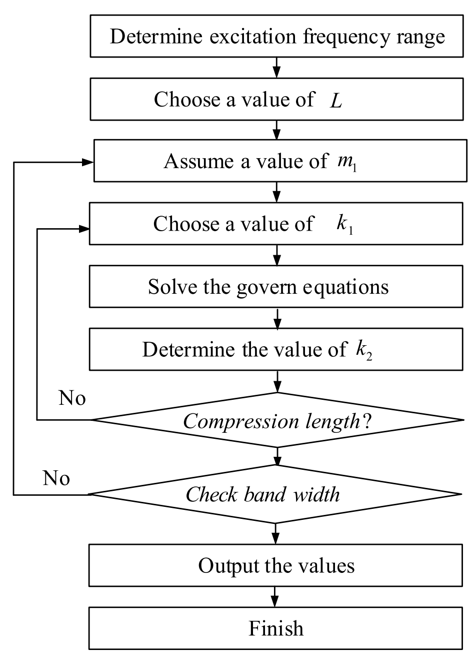

Fig. 12. Flowchart of the design process.

gap covered the pedestrian excitation frequency. Meanwhile, the mass of the attached IDVAs did not exceed 1% of the box girder weight. Thus, the parameters of the IDVAs were designed as follows: $k_1 = 1.95 \times 10^5$ N/m, $k_2 = 6.29 \times 10^4$ N/m, $m_1 = 59$ kg,

Table 2. Band gap of metamaterial beam with IDVAs

Unit cell, L (m)	Design parameters for IDVAs m_1 (kg)	k_1 (kN/m)	k_2 (kN/m)	b'	Band gap (Hz)	$\frac{Total\ mass\ of\ IDVAs}{Mass\ of\ box\ girder}$ (%)
1.5	48	105	84.2	0.51	1.09–1.87	1.12
2.0	59	195	62.9	0.95	0.88–3.30	0.98
2.5	178	271	96.7	1.05	0.92–2.82	2.50
3.0	242	278	106.9	0.82	0.98–2.66	2.83
3.5	316	332	132.8	0.65	1.15–2.42	3.17

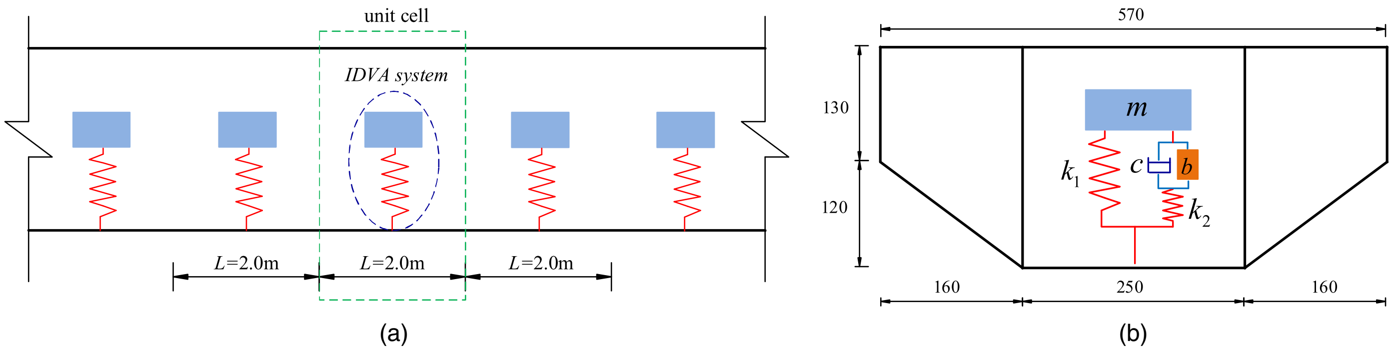

Fig. 13. Layout of the IDVAs: (a) perspective view; and (b) cross section.

$b' = 0.95$, and $\xi = 0$. The IDVAs were placed in the box girder every 2 m along the span of the footbridge, as shown in Fig. 13.

Three FEMs were built to study the vibration behavior of the simply supported steel footbridge with different vibration control measurements, including no control, MTMD as reported by Wang et al. (2019), and the proposed metamaterial. Four typical computational cases were designed, as listed in Table 3, to investigate the dynamic responses of the footbridges, including different crowd densities, human-bridge vibration forms (resonance and nonresonance), and different walking activities (walking and jumping). The pedestrian load was applied on the deck uniformly, where the total load under crowd excitation (Wang et al. 2019), $F_{crowd}(t)$, and jumping excitation (Bachmann and Ammann 1987), $F_{jump}(t)$, can be expressed as

$$F_{crowd}(t) = N_p G\left[1 + \sum_{i=1}^{4} c_i \sin(2\pi i f t + \phi_i)\right] \quad (22)$$

$$F_{jump}(t) = \alpha N_p G[\sin(2\pi i f t) + |\sin(2\pi i f t)|]/2 \quad (23)$$

in which, N_p = equivalent number of persons on the footbridge and can be determined by HIVOSS Project (Feldmann et al. 2008); G = average weight of a person, and assumed to be $G = 700$ N; c_i and ϕ_i, as determined by Kerr and Bishop (2001), are the dynamic factor and phase angle of the loading excitation of the ith order, respectively; f, ranging from 1.2 to 3.2 Hz, is the frequency of the pedestrian excitation; $\alpha = 1.5$ is the dynamic factor of the jumping load.

Table 3. Computational cases of pedestrian loading

Computational cases	Frequency (Hz)	Density (person/m^2)	Equivalent person number (person)	Description
1	1.56	1.50	94	Rhythmic crowd
2	2.95	1.50	94	Resonance generated
3	1.20	—	25	Slow jumping
4	2.95	—	25	Fast jumping

Figs. 14 and 15 show the dynamic response of the footbridge at midspan with different excitation frequencies. As can be seen, the vertical displacement of the footbridge for all cases is considerably lower than the predetermined limit value (the limit of vertical displacement for a footbridge is generally adopted as 1/1,000 of its span, i.e., 64 mm in this case). When the pedestrian frequency was 2.95 Hz (computational case 2), the peak vertical accelerations at midspan could be controlled effectively by the MTMD and metamaterials. Compared to the footbridge without control measurements (1.51 mm/s^2), the peak acceleration decreased by 85.4% for MTMD (0.22 mm/s^2) and 95.4% for metamaterials (0.07 mm/s^2). For computational case 1 (pedestrian frequency 1.56 Hz), the vibration attenuation efficiency of the metamaterial is observed to be superior to that of conventional MTMD, as shown in Table 4. It can be inferred that for the entire range of the pedestrian excitation frequency, the vibration reduction effectiveness of the metamaterial is better than that of MTMD, especially at $f \leq 1.56$ Hz. Similar conclusions can be drawn for slow jumping and fast jumping, as shown in Table 4 and Fig. 16.

Thus, it can be concluded that the proposed method can suppress the vibration response of the footbridge effectively for all loading cases when the excitation frequency lies in the band gap as the propagation of the vibration does not occur within the band gap. Moreover, the vibration reduction effectiveness depends only on the band gap, which can greatly simplify the design process for vibration attenuation. The influence of crowd distributions, crowd activities, stride rates, and time intervals of neighboring pedestrians passing through the footbridge can be neglected. We only require to design the band gap structure of the metamaterial according to the

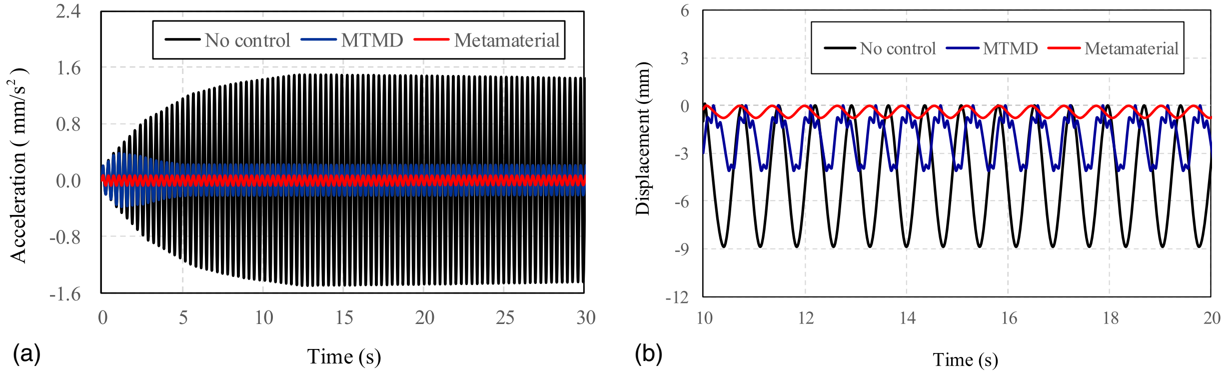

Fig. 14. Vibration response of the footbridge for $f = 2.95$ Hz: (a) vertical acceleration response; and (b) deformation.

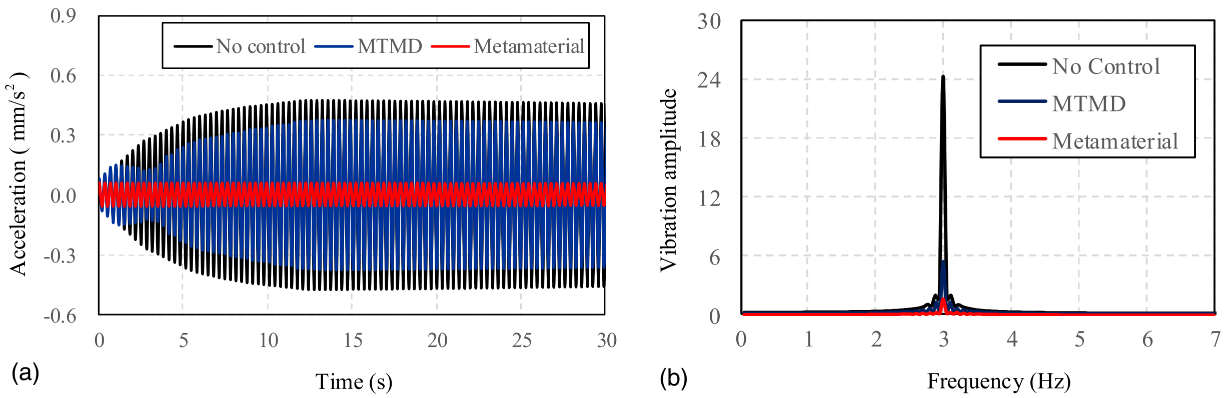

Fig. 15. Vibration response of the footbridge for $f = 1.56$ Hz: (a) vertical acceleration response; and (b) amplitude frequency curve.

Table 4. Peak vertical acceleration of the footbridge

Computational cases	Frequency (Hz)	Vertical acceleration ($\mathrm{mm/s^2}$) No control	MTMD	IDVAs	Reduction effectiveness (%) MTMD	IDVAs
1	1.56	0.48	0.37	0.03	22.9	93.7
2	2.95	1.51	0.22	0.07	85.4	95.4
3	1.20	0.41	0.27	0.02	34.1	95.1
4	2.95	1.03	0.32	0.04	68.9	96.1

eigenvalue equation and ensure that it covers the frequency range of external excitation.

Conclusions

A new technique is proposed as an alternative to control the vertical vibration of a simply supported footbridge. Unlike the traditional control measurements based on the structural dynamic theory, the proposed method is based on the theory of metamaterials. IDVAs were arranged periodically in the bottom slab to convert the

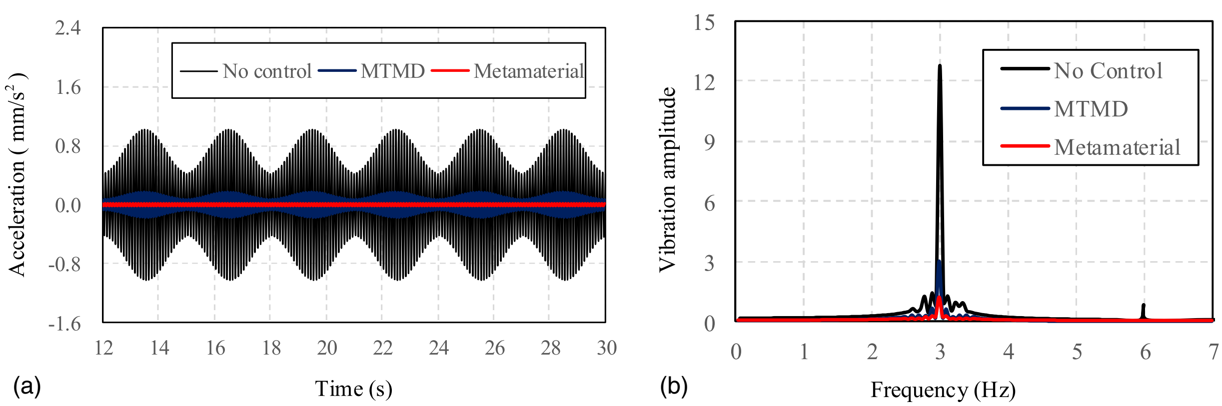

Fig. 16. Vibration response of the footbridge for fast jumping: (a) vertical acceleration response; and (b) amplitude frequency curve.

conventional box girder into a metamaterial beam with band gaps. The wave propagation constant and transmission curve were developed based on the SEM, and the band gap structure was designed to cover the frequency range of human excitation. The numerical results show that the vibration reduction effectiveness of the proposed method is better than that of MTMD in most cases. The following conclusions can be drawn:

1. A new vibration attenuation method for footbridges has been proposed based on the theory of band gap in metamaterials and validated by numerical experiments. The results show that the proposed method exhibits good performance in vibration attenuation. Moreover, human activities, crowd density, and crowd types have no effect on the vibration reduction effectiveness, which depends only on the band gap. This can greatly simplify the design process.
2. Design methods for the metamaterial beam with lightweight, low frequency, and wide band gap have been proposed. Low-frequency band gaps can be achieved with small mass resonators by the inertia amplification of IDVAs. Meanwhile, the width of the band gap can be broadened significantly by band gap merging.
3. Although the dispersive characteristics of vibration waves in long-span bridges are more complex than those of a simply supported beam, the proposed technique can be extended to control the vibration response of long-span bridges induced by vehicles, winds, and other excitations. Once the designed band gap covers the frequency range of excitations, the vibration responses of the structures can be suppressed significantly.

Data Availability Statement

Some or all data, models, or code that supports the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (Grant Nos. 51808208 and 51878151) and the Fundamental Research Funds for Central Universities (Grant No. 2242021R20011).

References

AFGC (Association Française de Génie Civil, Sétra). 2006. *Sétra: Evaluation du comportement vibratoire des passerelles piétonnes sous l'action des piétons*. Paris: AFGC.

Angelis, M. D., F. Petrini, and D. Pietrosanti. 2021. "Optimal design of the ideal grounded tuned mass damper inerter for comfort performances improvement in footbridges with practical implementation considerations." *Struct. Control Health Monit.* 28 (9): e28000. https://doi.org/10.1002/stc.2800.

Bachmann, H., and A. Ammann. 1987. *Vibrations in structures induced by man and machines*. Zurich, Switzerland: International Association for Bridges and Structural Engineering.

Banerjee, A., S. Adhikari, and M. Hussein. 2021. "Inertial amplification band-gap generation by coupling a levered mass with a locally resonant mass." *Int. J. Mech. Sci.* 207 (Oct): 106630. https://doi.org/10.1016/j.ijmecsci.2021.106630.

Caetano, E., A. Cunha, F. Magalhaes, and C. Moutinho. 2010. "Studies for controlling human-induced vibration of the Pedro e Ines footbridge, Portu-gal. Part 1: Assessment of dynamic behavior." *Eng. Struct.* 32 (4): 1069–1081. https://doi.org/10.1016/j.engstruct.2009.12.034.

CEN (European Committee for Standardisation). 2005. *Eurocode 5: Design of timber structures—Part 2: Bridges*. NBN EN 1995-2. Brussels, Belgium: CEN.

Cheng, Z. Q., and X. G. Hua. 2009. *Vibration and dynamic design of footbridge*. Beijing: China Communications Press.

Costa, M., D. A. Castello, C. Magluta, and N. Roitman. 2021. "On the optimal design and robustness of spatially distributed tuned mass dampers." *Mech. Syst. Sig. Process.* 150 (Mar): 107289. https://doi.org/10.1016/j.ymssp.2020.107289.

Dai, J., Z. D. Xu, and P. P. Gai. 2019. "Tuned mass-damper-inerter control of wind-induced vibration of flexible structures based on inerter location." *Eng. Struct.* 199 (15): 109585. https://doi.org/10.1016/j.engstruct.2019.109585.

Failla, G., R. Santoro, A. Burlon, and A. Russillo. 2020. "An exact approach to the dynamics of locally-resonant beams." *Mech. Res. Commun.* 103 (Jan): 103460. https://doi.org/10.1016/j.mechrescom.2019.103460.

Fang, X., K. Chuang, X. Jin, and Z. Huang. 2018. "Band-gap properties of elastic metamaterials with inerter-based dynamic vibration absorbers." *J. Appl. Mech.* 85 (7): 071010. https://doi.org/10.1115/1.4039898.

Feldmann, M., C. Heinemeyer, and M. Lukic. 2008. *HIVOSS—Human-induced vibrations of steel structures*. Brussels, Belgium: Office for Official Publications of the European Communities.

Frandsen, N. M., O. Bilal, J. Jensen, and M. Hussein. 2016. "Inertial amplification of continuous structures: Large band gaps from small masses." *J. Appl. Phys.* 119 (12): 124902. https://doi.org/10.1063/1.4944429.

Hu, Y., and M. Chen. 2015. "Performance evaluation for inerter-based dynamic vibration absorbers." *Int. J. Mech. Sci.* 99 (Aug): 297–307. https://doi.org/10.1016/j.ijmecsci.2015.06.003.

ISO. 2007 *Bases for design of structures—Serviceability of buildings and walkways against vibration*. ISO/DIS 10137. Geneva: ISO.

Jones, R., A. Pretlove, and R. Eyre. 1981. "Two case studies in the use of tuned vibration absorbers on footbridges." *Struct. Eng.* 59 (2): 27–32.

Kaveh, A., M. F. Farzam, and H. H. Jalali. 2020. "Statistical seismic performance assessment of tuned mass damper inerter." *Struct. Control Health Monit.* 27 (10): e2602. https://doi.org/10.1002/stc.2602.

Kerr, C., and M. Bishop. 2001. "Human induced loading on flexible staircases." *Eng. Struct.* 23 (1): 37–45. https://doi.org/10.1016/S0141-0296(00)00020-1.

Kulkarni, P., and J. Manimala. 2016. "Longitudinal elastic wave propagation characteristics of inertant acoustic metamaterials." *J. Appl. Phys.* 119 (24): 245101. https://doi.org/10.1063/1.4954074.

Kushwaha, M. S., P. Halevi, L. Dobrzynski, and B. Rouhani. 1993. "Acoustic band structure of periodic elastic composites." *Phys. Rev. Lett.* 71 (13): 2022–2025. https://doi.org/10.1103/PhysRevLett.71.2022.

Li, J., and S. Li. 2018. "Generating ultra wide low-frequency gap for transverse wave isolation via inertial amplification effects." *Phys. Lett. A* 382 (5): 241–247. https://doi.org/10.1016/j.physleta.2017.11.023.

Lievens, K., G. Lombaert, G. Roeck, and V. Peter. 2016. "Robust design of a TMD for the vibration serviceability of a footbridge." *Eng. Struct.* 123 (15): 408–418. https://doi.org/10.1016/j.engstruct.2016.05.028.

Liu, X., J. J. Zheng, T. Branislav, and H. J. Andrew. 2018. "Model identification methodology for fluid-based inerters." *Mech. Syst. Sig. Process.* 106 (Jun): 479–494. https://doi.org/10.1016/j.ymssp.2018.01.018.

Liu, Z., X. Zhang, Y. Mao, Y. Zhu, Z. Yang, and C. Chan. 2000. "Locally resonant sonic materials." *Science* 289 (5485): 1734. https://doi.org/10.1126/science.289.5485.1734.

Ma, G., and P. Sheng. 2016. "Acoustic metamaterials: From local resonances to broad horizons." *Sci. Adv.* 2 (2): e1501595. https://doi.org/10.1126/sciadv.1501595.

Ormondroyd, J., and D. Hartog. 1928. "Theory of the dynamic vibration absorber." *J. Appl. Mech.* 50: 9–22.

Petrini, F., A. Giarallis, and Z. Wang. 2020. "Optimal tuned mass-damper-inerter design in wind-excited tall buildings for occupants' comfort serviceability performance and energy harvesting." *Eng. Struct.* 204 (Feb): 109904. https://doi.org/10.1016/j.engstruct.2019.109904.

Pietrosanti, D., M. D. Angelis, and A. Giaralis. 2020. "Experimental study and numerical modeling of nonlinear dynamic response of SDOF system equipped with tuned mass damper inerter (TMDI) tested on shaking table under harmonic excitation." *Int. J. Mech. Sci.* 184 (Oct): 105762. https://doi.org/10.1016/j.ijmecsci.2020.105762.

Poovarodom, N., S. Kanchanosot, and P. Warnitchai. 2003. "Application of non-linear multiple tuned mass dampers to suppress man-induced vibrations of a pedestrian bridge." *Earthquake Eng. Struct. Dyn.* 32 (7): 1117–1131. https://doi.org/10.1002/eqe.265.

Shi, W., L. Wang, Z. Lu, and H. Wang. 2019. "Experimental and numerical study on adaptive-passive variable mass tuned mass damper." *J. Sound Vib.* 452 (Jul): 97–111. https://doi.org/10.1016/j.jsv.2019.04.008.

Smith, M. C. 2002. "Synthesis of mechanical networks: The inerter." *IEEE Trans. Autom. Contr.* 47 (10): 1648–1662. https://doi.org/10.1109/TAC.2002.803532.

Taflanidis, A. A., A. Giaralis, and D. Patsialis. 2019. "Multi-objective optimal design of inerter-based vibration absorbers for earthquake protection of multi-storey building structures." *J. Franklin Inst.* 356 (14): 7754–7784. https://doi.org/10.1016/j.jfranklin.2019.02.022.

Wang, D., C. Wu, Y. Zhang, and S. Li. 2019. "Study on vertical vibration control of long-span steel footbridge with tuned mass dampers under pedestrian excitation." *J. Constr. Steel Res.* 154 (Mar): 84–98. https://doi.org/10.1016/j.jcsr.2018.11.021.

Wen, Y., Z. Chen, and X. Hua. 2016. "Design and evaluation of tuned inerter-based dampers for the seismic control of MDOF structures." *J. Struct. Eng.* 143 (4): 04016207. https://doi.org/10.1061/(ASCE)ST.1943-541X.0001680.

Werkle, H., C. Butz, and R. Tatar. 2013. "Effectiveness of detuned TMD's for beam-like footbridges." *Adv. Struct. Eng.* 16 (1): 21–31. https://doi.org/10.1260/1369-4332.16.1.21.

Xiao, Y., J. Wen, and X. Wen. 2012. "Longitudinal wave band gaps in metamaterial-based elastic rods containing multi-degree-of-freedom resonators." *New J. Phys.* 14 (3): 033042. https://doi.org/10.1088/1367-2630/14/3/033042.

Yilmaz, C., and G. Hulbert. 2010. "Theory of phononic gaps induced by inertial amplification in finite structures." *Phys. Lett. A* 374 (34): 3576–3584. https://doi.org/10.1016/j.physleta.2010.07.001.

Yilmaz, C., G. Hulbert, and N. Kikuchi. 2007. "Phononic band gaps induced by inertial amplification in periodic media." *Phys. Rev. B* 76 (5): 054309. https://doi.org/10.1103/PhysRevB.76.054309.

Yuksel, O., and C. Yilmaz. 2020. "Realization of an ultrawide stop band in a 2-D elastic metamaterial with topologically optimized inertial amplification mechanisms." *Int. J. Solids Struct.* 203 (Oct): 138–150. https://doi.org/10.1016/j.ijsolstr.2020.07.018.

Zhu, Q. K., W. B. Yang, Y. F. Du, and S. Zivanovic. 2021. "Investigation of a vibration mitigation method based on crowd flow control on a footbridge." In Vol. 33 of *Structures*, 1495–1509. Amsterdam, Netherlands: Elsevier.

Zivanovic, S., A. Pavic, and P. Reynolds. 2005. "Vibration serviceability of footbridges under human-induced excitation: A literature review." *J. Sound Vib.* 279 (1–2): 1–74. https://doi.org/10.1016/j.jsv.2004.01.019.